Differential Count

The differential WBC count is the method in which the numbers of different types of white blood cells present in the blood are counted by examining a well stained peripheral blood smear (prepared by the above method). The number of each type of white blood cell is then expressed as a percentage of the total number of cells counted (Figure12.13).

Summary

Medical instruments help in correct diagnosis and treatment to observing patients, examining organs, making measurements and administering medication. Today's medical instruments are considerably more complicated and diverse, primarily because they incorporate electronic systems for sensing, transducing, manipulating, storing, and displaying data or information. The modern electronic stethoscope is a high precision instrument used to hear with clarity the internal sounds of the human body even in a noisy environment or through the heavy clothing of the patient. The sphygmomanometer is a device to diagnose pathological conditions such as hypertension and hypotension. A glucometer is a simple and portable medical device used to find out the approximate levels of blood glucose. Autoanalyser is a computer controlled device and used to quick estimation of various bio-chemical substances like glucose, urea, cholesterol, enzymes and other proteins present in body fluids.

Imaging instruments produce diagnostic images and help in the detection of the diseases or defects in an organ at a very early stage, when there are no clinical manifestations or the disease is undetected by the conventional methods of investigations. Then proper therapeutic or surgical measures can be adopted. An EEG is a test to evaluate the electrical activities of the brain. X-Rays are a form of electromagnetic radiation to visualize the internal structures of a patient. Ultrasound waves are used to image the foetus at different stages of pregnancy used to hear foetal heart sound, blood flow, diagnose the damages in heart and diagnosis of tumours, gall stones, kidney stones, and obstructions in the genital tracts.

Evaluation

1. The instrument used to measure blood pressure is
   a. Stethoscope
   b. Haemocytometer
   c. Sphygmomanometer
   d. Haemoglobinometer
2. Blood smear is used to study
   a. Total RBC
   b. Total WBC
   c. Haemoglobin
   d. Differential count of WBC
3. Diluting fluid used for total RBC count is
   a. Turk's solution
   b. Toisson solution
   c. Hayem's solution
   d. Folin phenol reagent
4. Normal diastolic blood pressure is
   a. 80 mmHg b. 100 mmHg
   c.120 mmHg d.140 mmHg
5. Foetus development can be observed using
   a. Ultra sonogram b. X-Ray
   c. ECG d. EEG

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